

Aim:

To create an Azure Load Testing resource and run a load test to evaluate the performance of a target endpoint and to create and demonstrate an Azure DevOps pipeline for automating application builds, tests, and deployment.

Load Testing

Azure Load Testing:

Azure Load Testing allows you to simulate high traffic and stress tests for your web applications and APIs to understand how they perform under load. It helps identify performance bottlenecks, scalability issues, and optimize resource usage before deployment.

Steps to Create an Azure Load Testing Resource:

Before you run your first test, you need to create the Azure Load Testing resource:

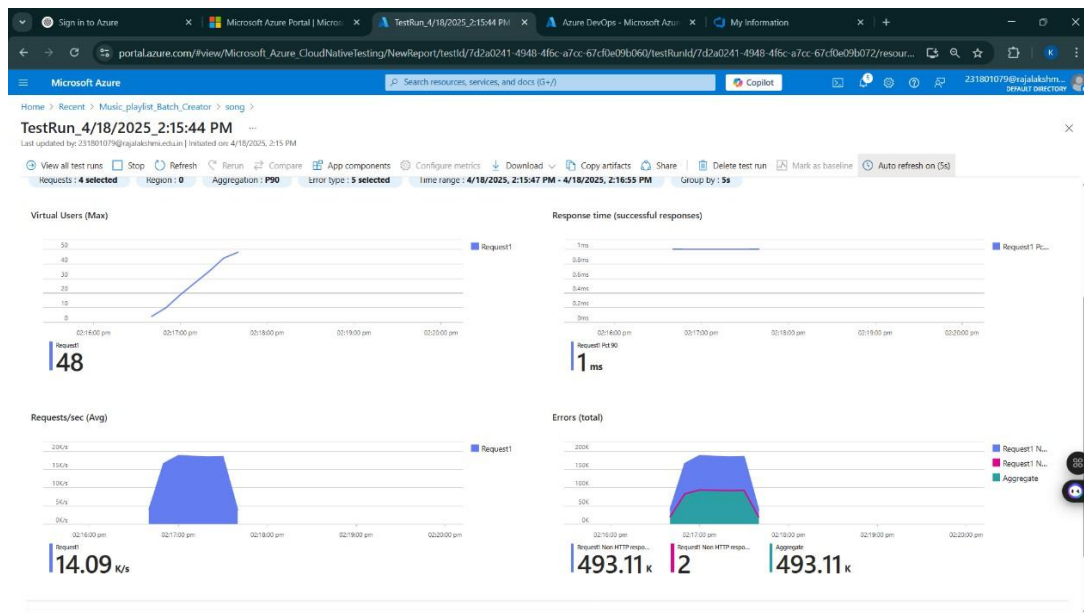
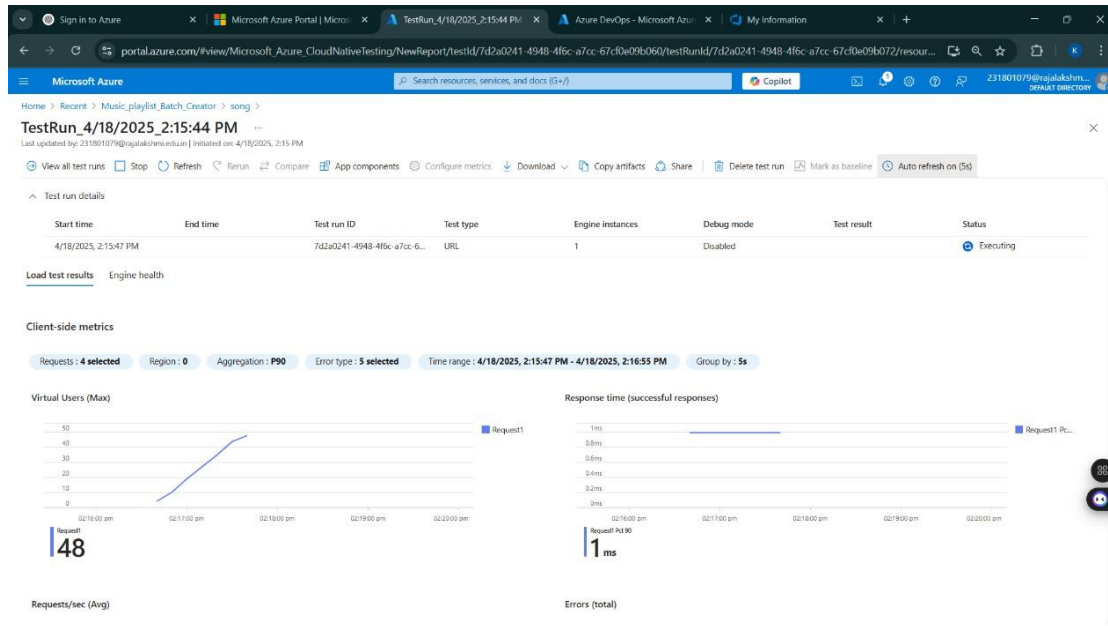
1. Sign in to Azure Portal
Go to <https://portal.azure.com> and log in.
2. Create the Resource
 - o Go to *Create a resource* → Search for “Azure Load Testing”.
 - o Select Azure Load Testing and click Create.
3. Fill in the Configuration Details
 - o *Subscription*: Choose your Azure subscription.
 - o *Resource Group*: Create new or select an existing one.
 - o *Name*: Provide a unique name (no special characters).
 - o *Location*: Choose the region for hosting the resource.
4. (Optional) Configure tags for categorization and billing.
5. Click Review + Create, then Create.
6. Once deployment is complete, click Go to resource.

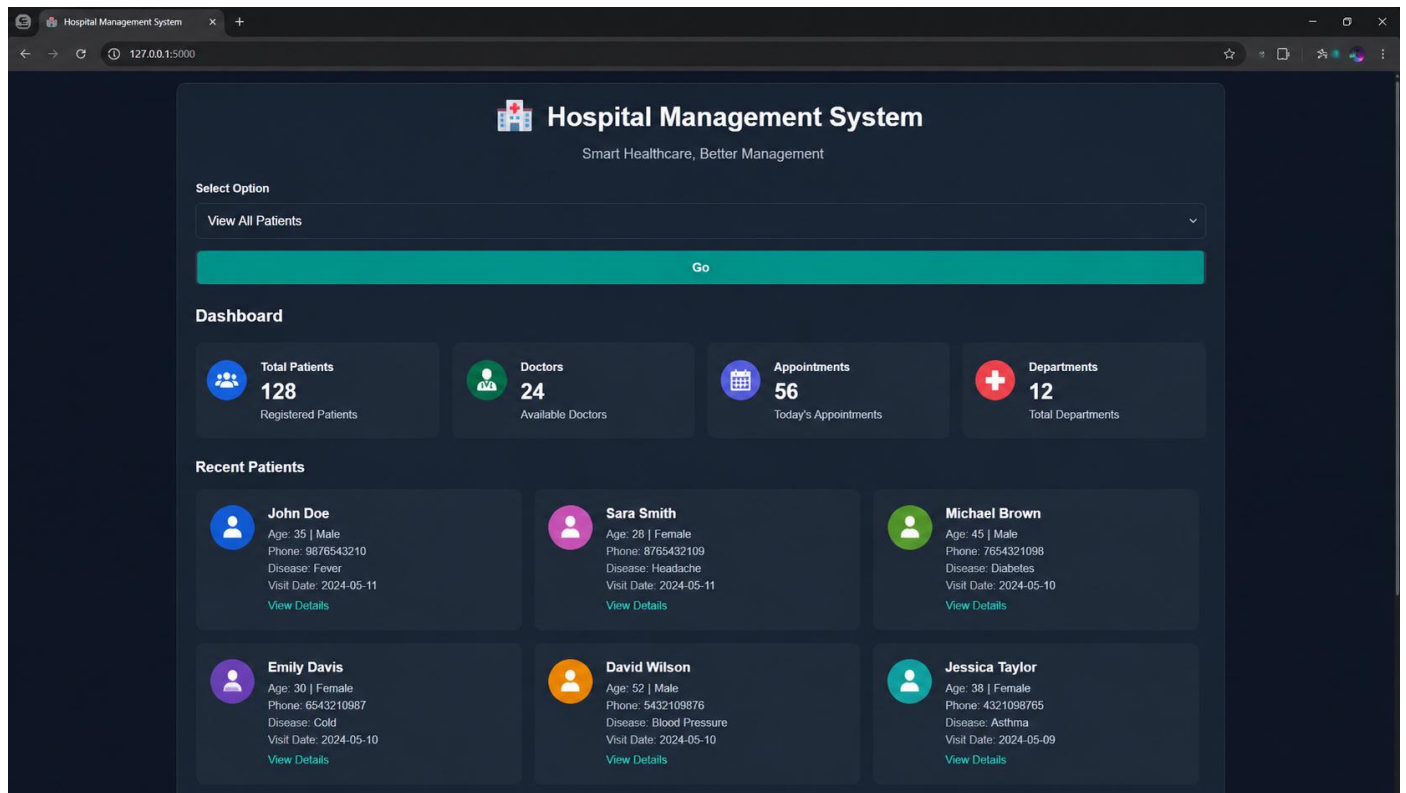
Steps to Create and Run a Load Test:

Once your resource is ready:

1. Go to your Azure Load Testing resource and click Add HTTP requests > Create.
2. Basics Tab
 - o *Test Name*: Provide a unique name.
 - o *Description*: (Optional) Add test purpose.
 - o *Run After Creation*: Keep checked.
3. Load Settings
 - o *Test URL*: Enter the target endpoint (e.g., <https://yourapi.com/products>).
4. Click Review + Create → Create to start the test.

Load Testing





Pipelines

Description:

This experiment demonstrates how to connect a GitHub-hosted Flask-based music recommendation project with Azure DevOps. The pipeline will automatically install dependencies, run basic tests, and publish artifacts. This ensures that every commit triggers checks for reliability and smooth deployment.

Steps:

1. Connect GitHub to Azure DevOps:
 - o In Azure DevOps, create a new project.
 - o Create a pipeline and select GitHub as the source.
 - o Authorize access to your GitHub repository, ensuring that Azure DevOps can pull the repository for your pipeline.
2. Create azure-pipelines.yml in Your Repo Root:
 - o In your GitHub repository, create a new file called azure-pipelines.yml in the root directory.
 - o Add the following basic pipeline configuration for Python and Flask:

yml Code

```
trigger:
- main

pool:
vmImage: 'ubuntu-latest'

steps:

- task: UsePythonVersion@0
inputs:
versionSpec: '3.x'
displayName: 'Set up Python'

- script: |
python -m pip install --upgrade pip
pip install flask
displayName: 'Install dependencies'

- script: |
python -c "print('Azure Pipeline Success')"
displayName: 'Run Python Script'
```

3. Pipeline Tasks Include:

- o Setting up the Python environment using the UsePythonVersion task.
- o Installing project dependencies from project/requirements.txt. Make sure the path to requirements.txt is correct (it is located under the project folder).
- o Running a simple Python script to verify that Python is set up correctly and the pipeline works.

4. Run and Monitor Pipeline:

- o Commit changes to the main branch of your repository to trigger the pipeline in Azure DevOps.
- o Monitor the logs in the Azure DevOps portal to view logs, errors, or success messages and ensure everything runs smoothly.

Pipeline

shikhaajith2024ids / Hospital Appointment Syste... / Pipelines / shikha34567.Azure_Hospital... / 20260512.13

Search

#20260512.13 • Update azure-pipelines.yml

shikha34567.Azure_Hospital_Management_Project

Run new

This run is being retained as one of 3 recent runs by pipeline. View retention leases

Summary Code Coverage

Individual CI by shikha34567

Yesterday at 10:47 PM 17s

Repository and version

shikha34567/Azure_Hospital_Management_Project

main a3de4822

Related

0 work items

0 published

Tests and coverage

Get started

View 3 changes Parameters

Stages Jobs

Stage

1 job completed 14s

Job 11s

Run Stage

Result:

Successfully created the Azure Load Testing resource and executed a load test to assess the performance of the specified endpoint and also demonstrated pipelines in azure devops.